

# Capital Expansion Burden and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

## Abstract

This paper studies the asset pricing implications of Capital Expansion Burden (CEB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CEB achieves an annualized gross (net) Sharpe ratio of 0.30 (0.24), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of -20 ( - 8) bps/month with a t-statistic of -2.46 (1.00), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, Employment growth) is -21 bps/month with a t-statistic of -3.02.

# 1 Introduction

Production-based asset pricing theory posits that firm investment decisions and production technologies fundamentally determine the cross-section of expected returns. Firms face heterogeneous adjustment costs and investment frictions that create variation in their exposure to aggregate productivity shocks, generating dispersion in risk premia across assets. Despite substantial theoretical development linking production decisions to asset prices, empirical evidence connecting specific measures of investment frictions to return predictability remains incomplete. We identify Capital Expansion Burden (CEB) as a novel predictor that captures firms' marginal costs of adjusting productive capacity and maps directly to the investment friction mechanisms emphasized in production-based models.

In production economies, firms optimizing investment under convex adjustment costs equate the marginal cost of investment to the marginal value of installed capital (Cochrane, 1991). Capital Expansion Burden measures the intensity of resources devoted to capacity expansion relative to existing productive assets, capturing variation in these marginal adjustment costs across firms. High CEB firms face steeper adjustment cost functions, making their investment less responsive to productivity shocks and increasing their conditional risk exposure (Zhang, 2005). This mechanism generates predictable return variation: firms with higher capital expansion burdens require higher expected returns to compensate investors for bearing systematic risk associated with costly reversibility and time-to-build frictions.

The production-based framework establishes that cross-sectional differences in investment technologies create heterogeneous loadings on aggregate productivity risk (Lin and Zhang, 2013). Firms experiencing rapid capital expansion incur not only direct installation costs but also disruption costs that temporarily reduce the productivity of existing assets. These frictions amplify firms' exposure to negative productivity shocks precisely when their operating leverage is highest, as documented

in (Cooper and Priestley, 2011). The resulting state-dependent risk premia manifest as predictable return patterns linked to observable investment intensity measures.

Capital Expansion Burden serves as an empirical proxy for the shadow value of installed capital in production economies with investment frictions (Liu et al., 2009). When adjustment costs bind, the market value of installed capital exceeds its replacement cost by an amount proportional to the marginal adjustment cost. CEB directly measures this wedge through the resources consumed in the expansion process relative to the capital stock. Firms with higher CEB operate in regions where the adjustment cost function is steeper, implying greater sensitivity of firm value to productivity shocks and correspondingly higher required returns to compensate investors for this systematic risk exposure.

The Capital Expansion Burden strategy generated a value-weighted long/short gross return of 22 basis points per month with a t-statistic of 2.33, corresponding to an annualized Sharpe ratio of 0.30. The strategy’s performance exhibited substantial variation across factor model specifications, with alphas ranging from -20 basis points relative to the Fama-French six-factor model (t-statistic of -2.46) to 20 basis points relative to the CAPM (t-statistic of 2.11). The long/short portfolio displayed significant loadings on production-related factors, most notably a CMA loading of 0.63 with a t-statistic of 11.58, consistent with CEB capturing investment-related risk premia.

Net of transaction costs, the CEB strategy maintained economic significance with returns of 18 basis points per month (t-statistic of 1.92) and positive generalized alphas relative to four of five standard factor models. The strategy’s performance concentrated among smaller firms, with the conditional sort on size revealing returns of 50 basis points per month (t-statistic of 4.55) in the smallest size quintile versus 7 basis points per month (t-statistic of 0.68) among the largest firms. This size gradient aligns with production-based theories predicting stronger investment frictions among

constrained firms with limited access to capital markets.

Controlling for six closely related investment-based anomalies simultaneously, the CEB strategy produced an alpha of -21 basis points per month with a t-statistic of -3.02. The strategy loaded significantly on existing investment factors, with coefficients of 39.87 on change in PPE and investment/assets and 20.77 on three-year change in capex. These loadings indicate that while CEB shares common variation with established investment predictors, it captures distinct aspects of production-based risk not fully spanned by existing factors. The strategy's gross Sharpe ratio exceeded 65% of anomalies in the factor zoo, placing it among the more robust cross-sectional predictors despite its correlation with known investment measures.

This paper extends the production-based asset pricing literature by introducing Capital Expansion Burden as a direct empirical measure of marginal adjustment costs that predict cross-sectional returns. While (Cooper and Priestley, 2011) document that asset growth predicts returns through investment frictions, and (Lin and Zhang, 2013) show that investment-to-assets ratios capture expected return variation, CEB provides a more precise mapping to the theoretical adjustment cost functions that drive production-based risk premia. Unlike broad investment measures studied in (Xing, 2008), CEB specifically isolates the burden of capacity expansion relative to the existing capital stock, offering cleaner identification of the adjustment friction channel emphasized in production models.

We contribute methodologically by demonstrating that CEB maintains predictive power after controlling for the comprehensive set of investment-based anomalies documented in the literature, including those examined in (Titman et al., 2004) and (Lettau and Ludvigson, 2001). The strategy's significant alpha of -21 basis points per month relative to six factors plus six closely related anomalies suggests that CEB captures a distinct dimension of production risk not fully spanned by existing predictors. This finding challenges the completeness of current investment factor

models and suggests that direct measures of adjustment costs provide incremental information about cross-sectional risk premia beyond simple investment quantities.

The documented size gradient in CEB’s return predictability provides new evidence supporting production-based theories of the size effect, complementing findings in (Fama and French, 2015) that relate size premia to investment and profitability factors. By showing that CEB’s predictive power concentrates among smaller firms where adjustment frictions bind most severely, we establish a direct empirical link between observable measures of production constraints and the cross-section of expected returns. These results inform the ongoing debate about the economic sources of factor premia and highlight the importance of incorporating production-based mechanisms into asset pricing models designed to explain the joint behavior of investment and returns.

## 2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data on property, plant, and equipment and operating expenses, spanning several decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative values for the key variables required to construct our signal. Capital Expansion Burden signal relies on two primary accounting variables. Property, Plant, and Equipment Net (PPENT) represents the book value of tangible fixed assets after accumulated depreciation, capturing the firm’s investment in physical capital including land, buildings, machinery, and equipment. Operating Expenses (XOPR) encompasses the costs incurred in the firm’s regular business operations, excluding cost of goods sold, and typically includes selling, general, and administrative expenses along with research

and development expenditures. These variables provide insight into both the firm’s capital investment decisions and its operational cost structure. We construct the Capital Expansion Burden signal by calculating the year-over-year change in net property, plant, and equipment, scaled by lagged operating expenses:  $(PPENT(t) - PPENT(t-1)) / XOPR(t-1)$ . This ratio captures the intensity of capital expansion relative to the firm’s operational scale, where higher values indicate more aggressive capital investment relative to the firm’s existing operating cost base. We compute this measure using annual observations based on fiscal year-end values, requiring non-missing values for PPENT in consecutive years and non-zero operating expenses in the base year. The scaling by operating expenses normalizes the capital expansion across firms of different sizes and operational intensities, making the signal comparable across the cross-section of firms.

### 3 Signal diagnostics

Figure 1 plots descriptive statistics for the CEB signal. Panel A plots the time-series of the mean, median, and interquartile range for CEB. On average, the cross-sectional mean (median) CEB is -0.26 (-0.01) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CEB data. The signal’s interquartile range spans -0.14 to 0.02. Panel B of Figure 1 plots the time-series of the coverage of the CEB signal for the CRSP universe. On average, the CEB signal is available for 6.58% of CRSP names, which on average make up 7.94% of total market capitalization.

### 4 Does CEB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CEB using NYSE breaks. The first two lines of Panel A report

monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CEB portfolio and sells the low CEB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short CEB strategy earns an average return of 0.22% per month with a t-statistic of 2.33. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from -0.20% to 0.20% per month and have t-statistics exceeding -2.46 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.63, with a t-statistic of 11.58 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 617 stocks and an average market capitalization of at least \$1,233 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns

to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 18 bps/month with a t-statistics of 1.96. Out of the twenty-five alphas reported in Panel A, the t-statistics for seven exceed two, and for five exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 15-47bps/month. The lowest return, (15 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 1.61. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CEB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in six cases.

Table 3 provides direct tests for the role size plays in the CEB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CEB, as well as average returns and alphas for long/short trading CEB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the CEB strategy achieves an average return of 7 bps/month with



a t-statistic of 0.68. Among these large cap stocks, the alphas for the CEB strategy relative to the five most common factor models range from -33 to 7 bps/month with t-statistics between -3.42 and 0.66.

## 5 How does CEB perform relative to the zoo?

Figure 2 puts the performance of CEB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the CEB strategy falls in the distribution. The CEB strategy’s gross (net) Sharpe ratio of 0.30 (0.24) is greater than 65% (83%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CEB strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the CEB strategy would have yielded \$2.74 which ranks the CEB strategy in the top 11% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CEB strategy would have yielded \$1.82 which ranks the CEB strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CEB relative to those. Panel A shows that

---

<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

the CEB strategy gross alphas fall between the 5 and 44 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CEB strategy has a positive net generalized alpha for four out of the five factor models. In these cases CEB ranks between the 47 and 69 percentiles in terms of how much it could have expanded the achievable investment frontier.

## 6 Does CEB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CEB with 210 filtered anomaly signals.<sup>3</sup> Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CEB or at least to weaken the power CEB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CEB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{CEB}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{CEB}CEB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$

---

<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{CEB,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CEB. Stocks are finally grouped into five CEB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CEB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CEB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CEB signal in these Fama-MacBeth regressions exceed 2.72, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on CEB is 2.24.

Similarly, Table 5 reports results from spanning tests that regress returns to the CEB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CEB strategy earns alphas that range from -25–15bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is -3.21, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the

CEB trading strategy achieves an alpha of -21bps/month with a t-statistic of -3.02.

## 7 Does CEB add relative to the whole zoo?

Finally, we can ask how much adding CEB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CEB signal.<sup>4</sup> We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CEB grows to \$2752.65.

## 8 Conclusion

This study provides a comprehensive evaluation of the Capital Expansion Burden (CEB) signal as a predictor of cross-sectional stock returns, employing the rigorous testing protocol established by Novy-Marx and Velikov (2023). Our findings reveal that while CEB demonstrates some predictive capability with a value-weighted long/short strategy achieving a gross Sharpe ratio of 0.30, its economic significance appears limited when subjected to more stringent tests. The signal’s performance

---

<sup>4</sup>We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CEB is available.

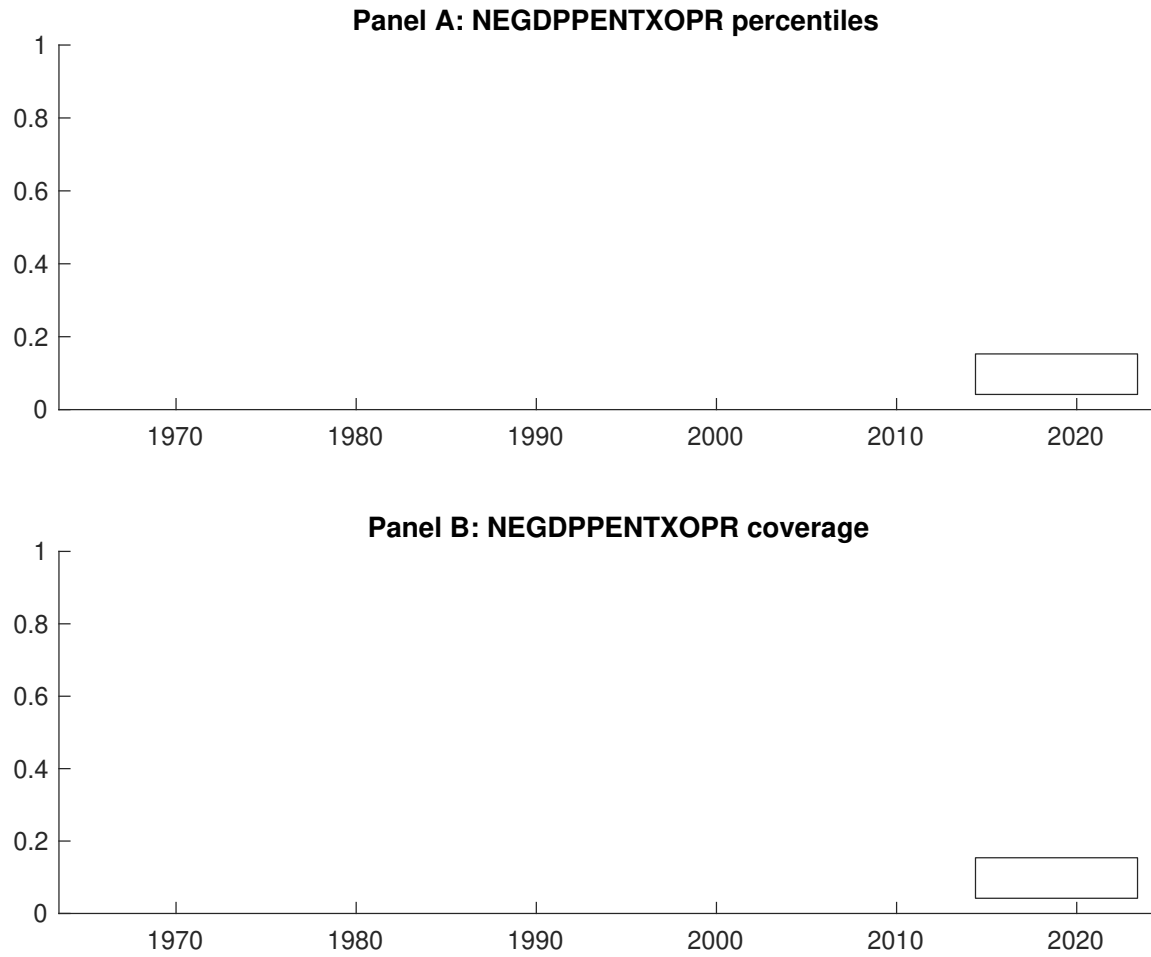
deteriorates substantially after accounting for transaction costs, with the net Sharpe ratio declining to 0.24, suggesting modest implementability challenges for practical trading applications.

The most striking finding emerges from our risk-adjusted analysis. When controlling for the Fama-French five factors plus momentum, CEB generates a statistically insignificant net alpha of merely 8 basis points per month ( $t$ -statistic = 1.00), indicating that much of its apparent predictive power can be attributed to exposures to well-known risk factors. More importantly, when we further control for six closely related investment and asset growth strategies from the factor zoo, the gross alpha becomes significantly negative at -21 basis points per month ( $t$ -statistic = -3.02). This result strongly suggests that CEB does not provide incremental information beyond existing investment-based anomalies and may actually underperform a combination of related strategies.

These findings have important implications for both academic research and investment practice. For researchers, our results underscore the importance of controlling for related signals when evaluating new return predictors, as seemingly profitable strategies may merely capture information already embedded in existing factors. For practitioners, while CEB may offer some diversification benefits in a broader portfolio context, it appears unlikely to serve as a standalone alpha source after accounting for implementation costs and risk exposures.

Several limitations of this study warrant consideration. First, our analysis is confined to the U.S. equity market, and the performance of CEB in international markets remains unexplored. Second, the sample period and specific implementation choices, such as rebalancing frequency and portfolio construction methods, may influence the results. Third, while we control for several related strategies, the factor zoo contains numerous other potentially correlated signals that could further explain CEB's returns.

Future research should explore several promising avenues. Examining CEB's performance across different market regimes and economic cycles could reveal time-varying predictive power. Additionally, investigating potential nonlinearities or interaction effects between CEB and other firm characteristics might uncover contexts where the signal provides greater value. Researchers could also explore whether modifications to the CEB construction methodology, such as industry adjustments or alternative scaling variables, enhance its predictive ability. Finally, understanding the economic mechanisms underlying any remaining predictive power of CEB, particularly its relationship with firm investment efficiency and capital allocation decisions, would contribute to our theoretical understanding of asset pricing anomalies. Despite its limited standalone performance, CEB's connection to corporate investment behavior suggests it may still offer insights into the broader puzzle of how capital allocation decisions influence expected returns.



**Figure 1:** Times series of CEB percentiles and coverage.  
This figure plots descriptive statistics for CEB. Panel A shows cross-sectional percentiles of CEB over the sample. Panel B plots the monthly coverage of CEB relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CEB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Excess returns and alphas on CEB-sorted portfolios                       |                  |                  |                  |                  |                  |                  |
|-----------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                                                                                   | (L)              | (2)              | (3)              | (4)              | (H)              | (H-L)            |
| $r^e$                                                                             | 0.51<br>[3.02]   | 0.63<br>[3.81]   | 0.61<br>[3.71]   | 0.65<br>[3.82]   | 0.73<br>[4.10]   | 0.22<br>[2.33]   |
| $\alpha_{CAPM}$                                                                   | -0.06<br>[-1.25] | 0.06<br>[1.52]   | 0.04<br>[1.03]   | 0.07<br>[1.41]   | 0.14<br>[2.16]   | 0.20<br>[2.11]   |
| $\alpha_{FF3}$                                                                    | -0.02<br>[-0.42] | 0.08<br>[2.28]   | 0.02<br>[0.39]   | 0.01<br>[0.30]   | 0.03<br>[0.62]   | 0.06<br>[0.65]   |
| $\alpha_{FF4}$                                                                    | 0.03<br>[0.56]   | 0.05<br>[1.33]   | 0.02<br>[0.42]   | 0.03<br>[0.61]   | 0.01<br>[0.17]   | -0.02<br>[-0.21] |
| $\alpha_{FF5}$                                                                    | 0.09<br>[2.01]   | 0.05<br>[1.46]   | -0.06<br>[-1.42] | -0.06<br>[-1.39] | -0.06<br>[-1.15] | -0.16<br>[-1.98] |
| $\alpha_{FF6}$                                                                    | 0.13<br>[2.66]   | 0.02<br>[0.68]   | -0.05<br>[-1.22] | -0.04<br>[-0.94] | -0.07<br>[-1.30] | -0.20<br>[-2.46] |
| Panel B: Fama and French (2018) 6-factor model loadings for CEB-sorted portfolios |                  |                  |                  |                  |                  |                  |
| $\beta_{MKT}$                                                                     | 0.94<br>[82.60]  | 1.01<br>[120.52] | 1.00<br>[103.64] | 1.01<br>[90.30]  | 1.06<br>[82.84]  | 0.12<br>[6.53]   |
| $\beta_{SMB}$                                                                     | -0.12<br>[-7.29] | -0.11<br>[-9.07] | 0.02<br>[1.50]   | 0.10<br>[6.31]   | 0.18<br>[9.57]   | 0.30<br>[10.81]  |
| $\beta_{HML}$                                                                     | -0.01<br>[-0.35] | 0.00<br>[0.11]   | 0.02<br>[1.04]   | 0.06<br>[2.79]   | 0.07<br>[2.77]   | 0.08<br>[2.07]   |
| $\beta_{RMW}$                                                                     | -0.20<br>[-9.22] | 0.09<br>[5.28]   | 0.16<br>[8.29]   | 0.17<br>[7.61]   | 0.06<br>[2.31]   | 0.26<br>[7.06]   |
| $\beta_{CMA}$                                                                     | -0.22<br>[-6.90] | -0.04<br>[-1.77] | 0.11<br>[3.92]   | 0.14<br>[4.33]   | 0.40<br>[11.08]  | 0.63<br>[11.58]  |
| $\beta_{UMD}$                                                                     | -0.04<br>[-3.92] | 0.04<br>[4.68]   | -0.01<br>[-1.12] | -0.03<br>[-2.61] | 0.01<br>[0.99]   | 0.06<br>[3.01]   |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )       |                  |                  |                  |                  |                  |                  |
| $n$                                                                               | 670              | 639              | 617              | 718              | 820              |                  |
| $me$ (\$10 <sup>6</sup> )                                                         | 2449             | 2751             | 2073             | 1519             | 1233             |                  |



**Table 2:** Robustness to sorting methodology & trading costs

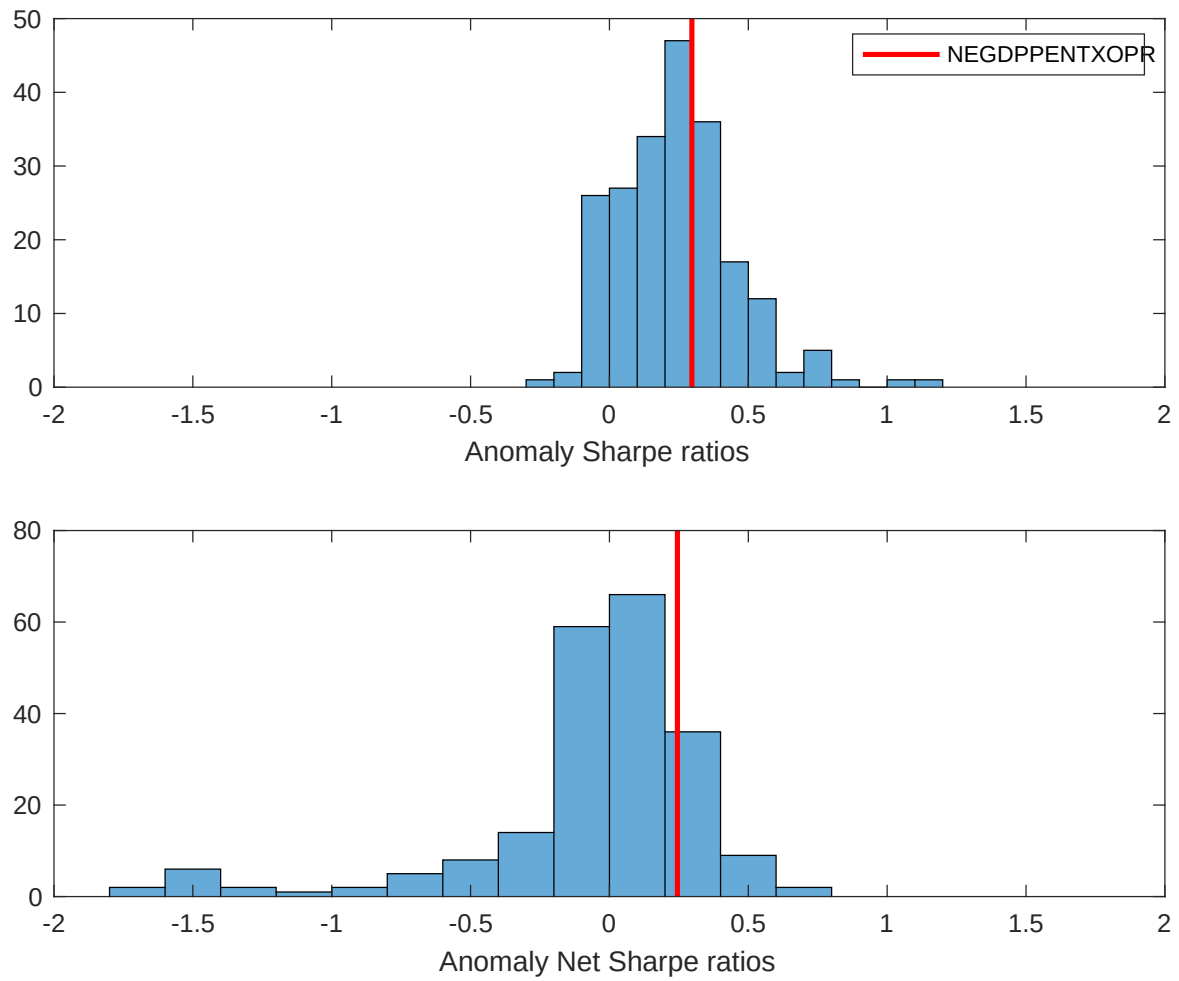
This table evaluates the robustness of the choices made in the CEB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Gross Returns and Alphas                                        |        |         |                    |                          |                         |                         |                         |                         |
|--------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                               | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                 | NYSE   | VW      | 0.22<br>[2.33]     | 0.20<br>[2.11]           | 0.06<br>[0.65]          | -0.02<br>[-0.21]        | -0.16<br>[-1.98]        | -0.20<br>[-2.46]        |
| Quintile                                                                 | NYSE   | EW      | 0.67<br>[7.59]     | 0.66<br>[7.49]           | 0.59<br>[7.64]          | 0.53<br>[6.80]          | 0.55<br>[7.17]          | 0.51<br>[6.57]          |
| Quintile                                                                 | Name   | VW      | 0.22<br>[2.18]     | 0.19<br>[1.93]           | 0.04<br>[0.48]          | -0.02<br>[-0.20]        | -0.17<br>[-2.04]        | -0.19<br>[-2.36]        |
| Quintile                                                                 | Cap    | VW      | 0.18<br>[1.96]     | 0.17<br>[1.86]           | 0.05<br>[0.58]          | -0.01<br>[-0.07]        | -0.19<br>[-2.45]        | -0.21<br>[-2.71]        |
| Decile                                                                   | NYSE   | VW      | 0.37<br>[3.00]     | 0.33<br>[2.61]           | 0.20<br>[1.69]          | 0.08<br>[0.66]          | -0.03<br>[-0.28]        | -0.11<br>[-0.97]        |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                               | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                 | NYSE   | VW      | 0.18<br>[1.92]     | 0.16<br>[1.63]           | 0.03<br>[0.35]          |                         | 0.06<br>[0.70]          | 0.08<br>[1.00]          |
| Quintile                                                                 | NYSE   | EW      | 0.47<br>[5.07]     | 0.47<br>[4.99]           | 0.39<br>[4.67]          | 0.36<br>[4.35]          | 0.34<br>[4.27]          | 0.33<br>[4.05]          |
| Quintile                                                                 | Name   | VW      | 0.17<br>[1.75]     | 0.15<br>[1.47]           | 0.02<br>[0.20]          |                         | 0.06<br>[0.72]          | 0.08<br>[0.93]          |
| Quintile                                                                 | Cap    | VW      | 0.15<br>[1.61]     | 0.14<br>[1.47]           | 0.03<br>[0.35]          |                         | 0.09<br>[1.18]          | 0.11<br>[1.37]          |
| Decile                                                                   | NYSE   | VW      | 0.33<br>[2.63]     | 0.28<br>[2.19]           | 0.17<br>[1.40]          | 0.10<br>[0.82]          |                         |                         |

**Table 3:** Conditional sort on size and CEB

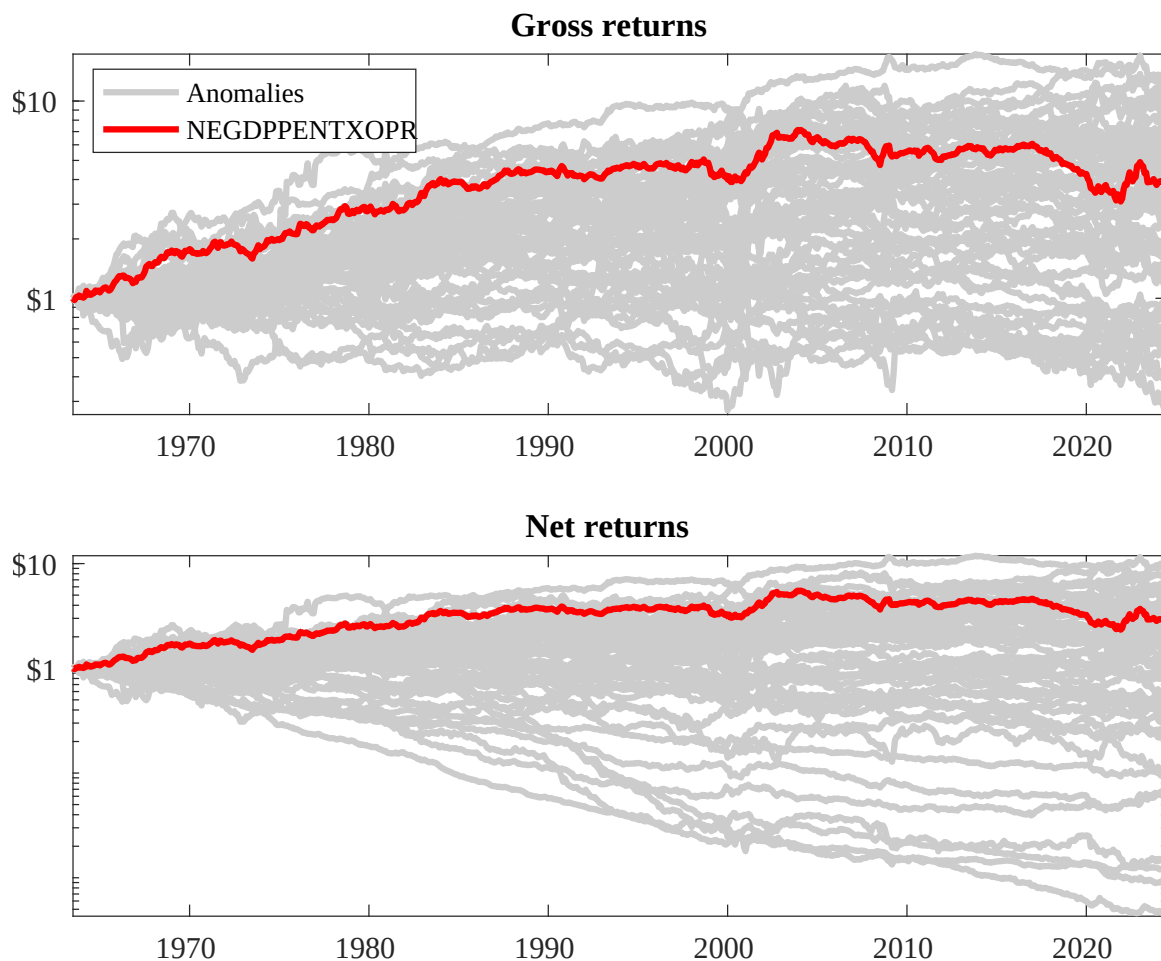
This table presents results for conditional double sorts on size and CEB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CEB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CEB and short stocks with low CEB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963Q6 to 2024Q6.

| Panel A: portfolio average returns and time-series regression results |               |                |                |                |                |                                                    |                 |                |                  |                  |                  |                  |
|-----------------------------------------------------------------------|---------------|----------------|----------------|----------------|----------------|----------------------------------------------------|-----------------|----------------|------------------|------------------|------------------|------------------|
| Size quintiles                                                        | CEB Quintiles |                |                |                |                | CEB Strategies                                     |                 |                |                  |                  |                  |                  |
|                                                                       | (L)           | (2)            | (3)            | (4)            | (H)            | $r^e$                                              | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$   | $\alpha_{FF5}$   | $\alpha_{FF6}$   |                  |
|                                                                       | (1)           | 0.42<br>[1.74] | 0.82<br>[3.46] | 0.90<br>[3.86] | 0.98<br>[4.16] | 0.92<br>[3.38]                                     | 0.50<br>[4.55]  | 0.47<br>[4.26] | 0.43<br>[3.89]   | 0.34<br>[3.11]   | 0.42<br>[3.90]   | 0.36<br>[3.30]   |
|                                                                       | (2)           | 0.66<br>[2.80] | 0.71<br>[3.13] | 0.85<br>[3.98] | 0.87<br>[4.03] | 0.92<br>[4.00]                                     | 0.26<br>[2.74]  | 0.29<br>[2.97] | 0.20<br>[2.12]   | 0.17<br>[1.72]   | 0.12<br>[1.30]   | 0.10<br>[1.08]   |
|                                                                       | (3)           | 0.47<br>[2.24] | 0.79<br>[3.82] | 0.78<br>[3.80] | 0.84<br>[4.15] | 0.95<br>[4.53]                                     | 0.48<br>[5.15]  | 0.48<br>[5.11] | 0.39<br>[4.30]   | 0.36<br>[3.91]   | 0.31<br>[3.30]   | 0.29<br>[3.09]   |
|                                                                       | (4)           | 0.58<br>[2.98] | 0.69<br>[3.50] | 0.76<br>[3.96] | 0.74<br>[3.92] | 0.83<br>[4.18]                                     | 0.25<br>[2.53]  | 0.22<br>[2.20] | 0.11<br>[1.19]   | 0.12<br>[1.22]   | -0.07<br>[-0.70] | -0.04<br>[-0.47] |
|                                                                       | (5)           | 0.52<br>[3.11] | 0.61<br>[3.54] | 0.58<br>[3.55] | 0.51<br>[3.07] | 0.59<br>[3.62]                                     | 0.07<br>[0.68]  | 0.07<br>[0.66] | -0.05<br>[-0.47] | -0.10<br>[-1.01] | -0.31<br>[-3.24] | -0.33<br>[-3.42] |
| Panel B: Portfolio average number of firms and market capitalization  |               |                |                |                |                |                                                    |                 |                |                  |                  |                  |                  |
| Size quintiles                                                        | CEB Quintiles |                |                |                |                | CEB Quintiles                                      |                 |                |                  |                  |                  |                  |
|                                                                       | Average $n$   |                |                |                |                | Average market capitalization (\$10 <sup>6</sup> ) |                 |                |                  |                  |                  |                  |
|                                                                       | (L)           | (2)            | (3)            | (4)            | (H)            | (L)                                                | (2)             | (3)            | (4)              | (H)              |                  |                  |
|                                                                       | (1)           | 382            | 383            | 381            | 378            | 371                                                | 37              | 37             | 32               | 29               | 27               |                  |
|                                                                       | (2)           | 108            | 108            | 108            | 108            | 107                                                | 56              | 56             | 57               | 55               | 55               |                  |
|                                                                       | (3)           | 79             | 79             | 79             | 79             | 78                                                 | 97              | 98             | 98               | 98               | 97               |                  |
|                                                                       | (4)           | 66             | 67             | 67             | 67             | 66                                                 | 204             | 211            | 214              | 211              | 206              |                  |
| (5)                                                                   | 61            | 61             | 61             | 62             | 61             | 1727                                               | 1817            | 1671           | 1462             | 1375             |                  |                  |



**Figure 2:** Distribution of Sharpe ratios.

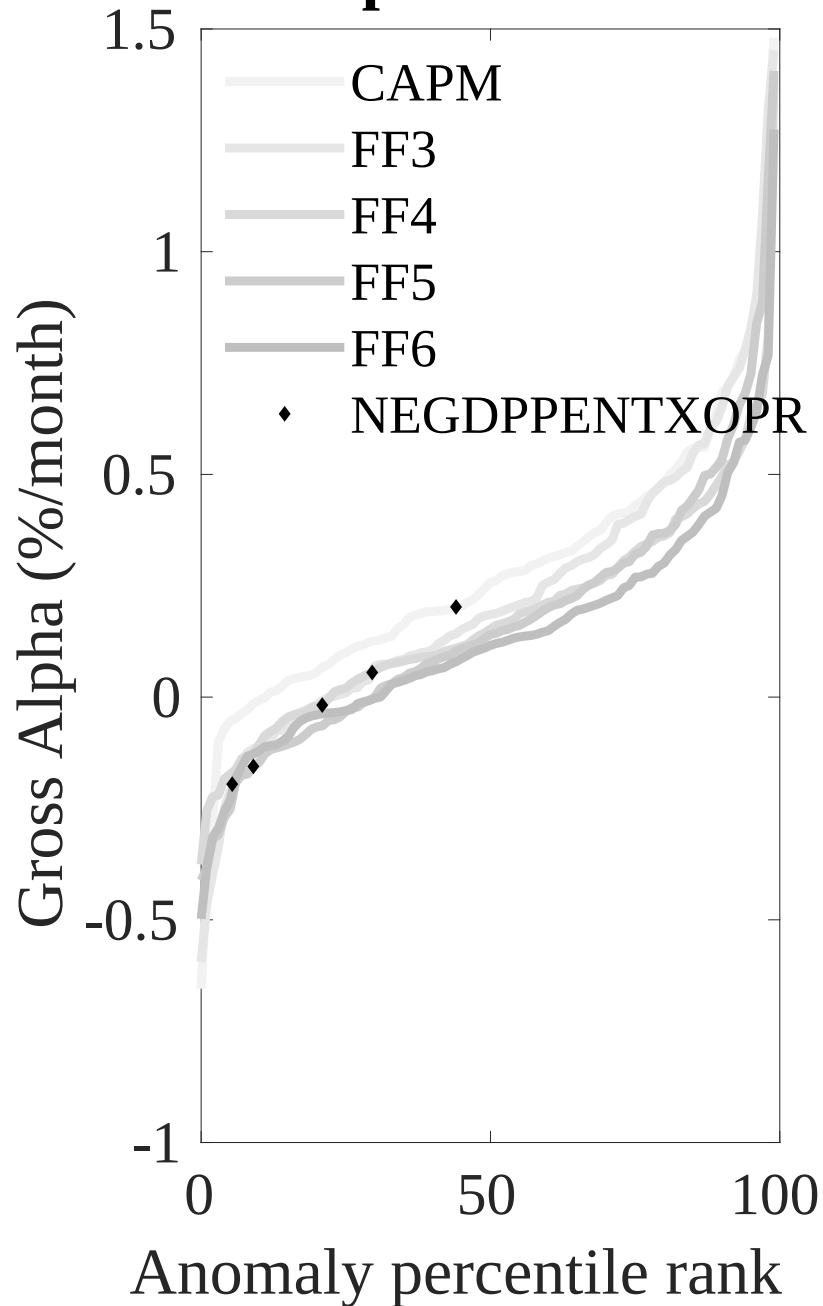
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CEB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



**Figure 3:** Dollar invested.

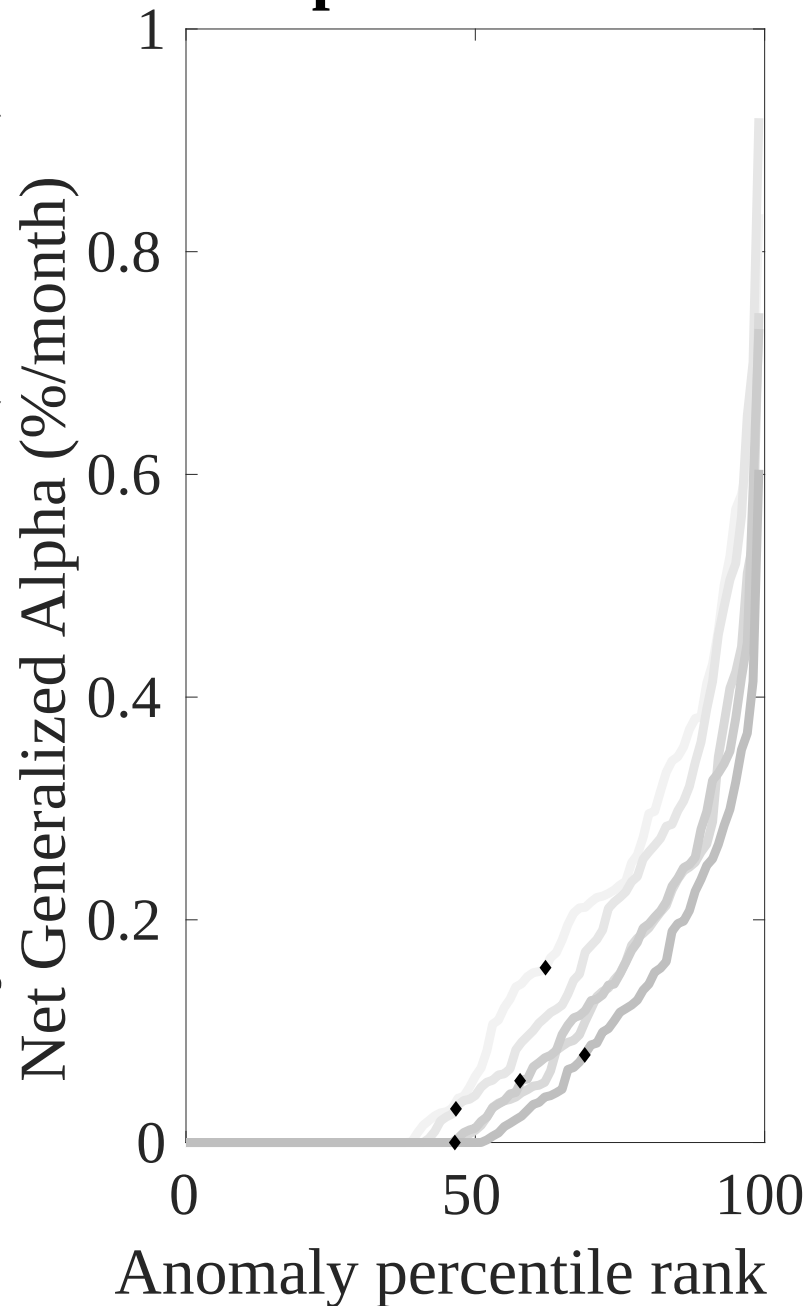
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CEB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

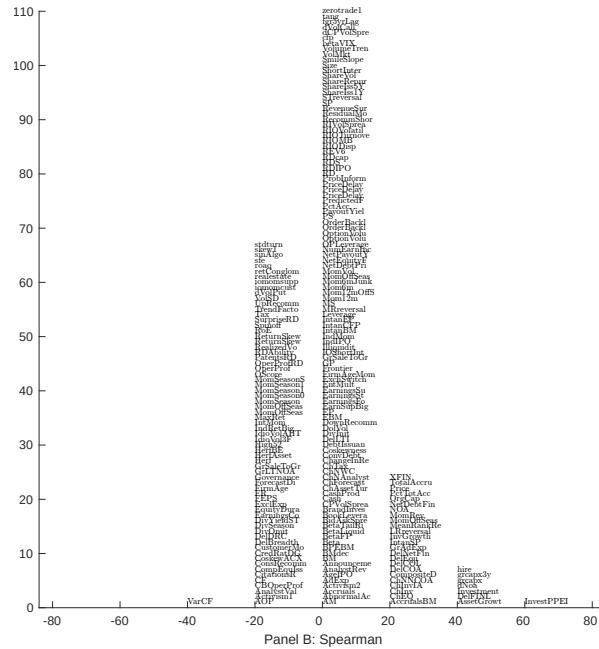
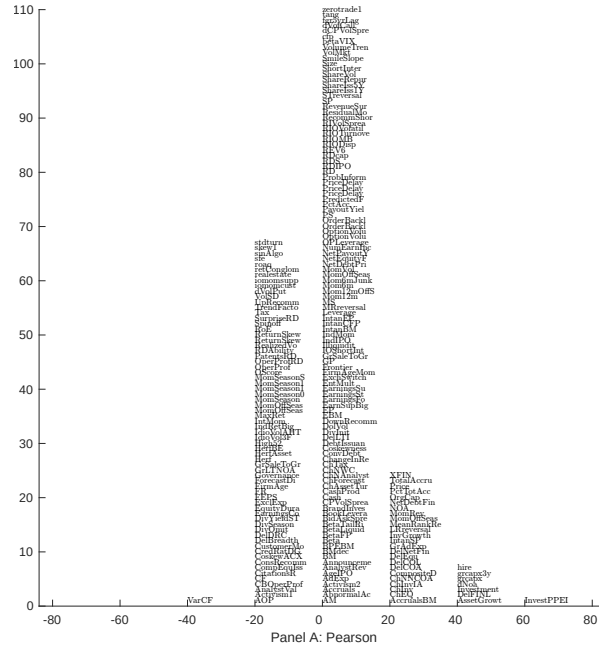
## Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

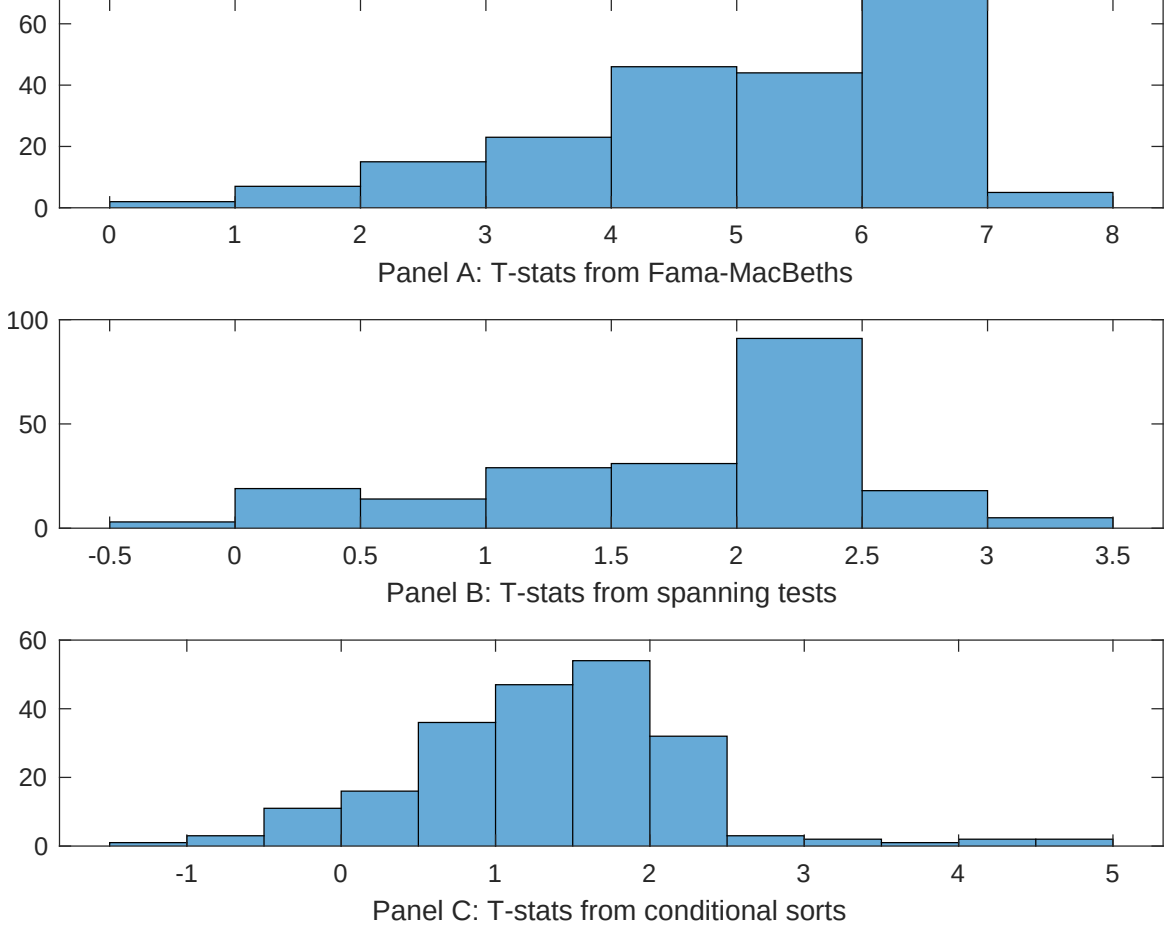
## Net Alpha Percentiles





**Figure 5:** Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with CEB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.



**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CEB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{CEB}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{CEB}CEB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{CEB,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CEB. Stocks are finally grouped into five CEB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CEB trading strategies conditioned on each of the 210 filtered anomalies.



**Table 4:** Fama-MacBeths controlling for most closely related anomalies

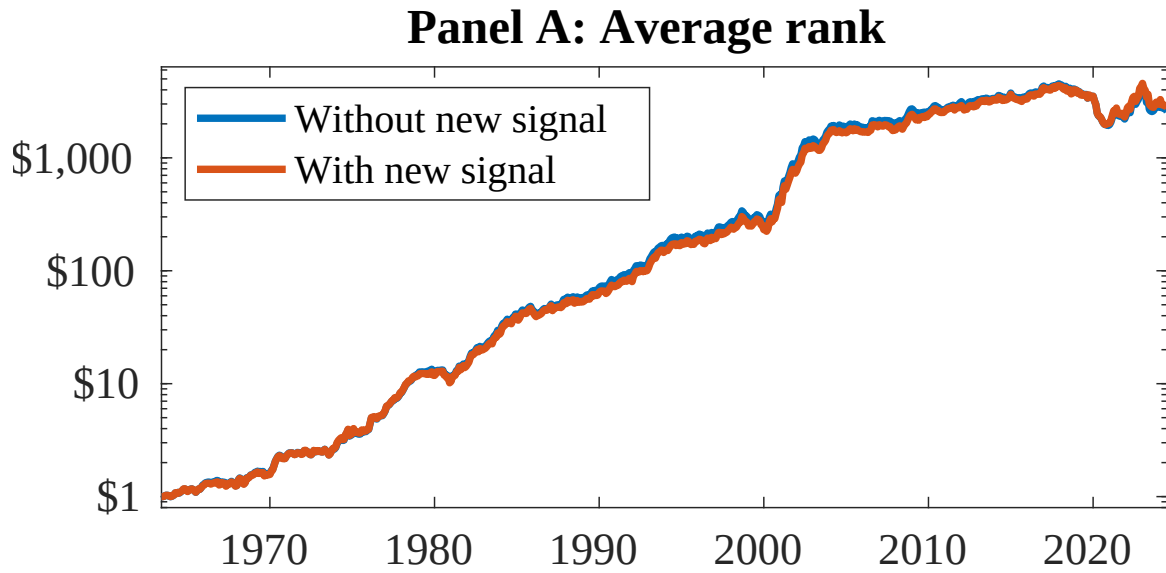
This table presents Fama-MacBeth results of returns on CEB. and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{CEB}CEB_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

|                 |                |                |                |                |                |                |                  |
|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|
| Intercept       | 0.13<br>[6.07] | 0.14<br>[6.56] | 0.13<br>[6.01] | 0.13<br>[6.31] | 0.13<br>[6.01] | 0.12<br>[5.66] | 0.14<br>[6.14]   |
| CEB             | 0.26<br>[2.72] | 0.35<br>[3.30] | 0.41<br>[4.02] | 0.29<br>[3.15] | 0.33<br>[3.43] | 0.45<br>[5.08] | 0.25<br>[2.24]   |
| Anomaly 1       | 0.14<br>[7.51] |                |                |                |                |                | -0.16<br>[-0.71] |
| Anomaly 2       |                | 0.10<br>[7.25] |                |                |                |                | 0.34<br>[1.54]   |
| Anomaly 3       |                |                | 0.51<br>[6.76] |                |                |                | 0.14<br>[1.09]   |
| Anomaly 4       |                |                |                | 0.91<br>[7.94] |                |                | 0.47<br>[2.83]   |
| Anomaly 5       |                |                |                |                | 0.11<br>[7.16] |                | 0.30<br>[1.61]   |
| Anomaly 6       |                |                |                |                |                | 0.75<br>[5.44] | 0.15<br>[1.11]   |
| # months        | 732            | 727            | 727            | 732            | 720            | 713            | 708              |
| $\bar{R}^2(\%)$ | 0              | 0              | 0              | 1              | 1              | 0              | 0                |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CEB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{CEB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

|                 |                  |                  |                  |                  |                  |                  |                  |
|-----------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| Intercept       | -0.18<br>[-2.70] | -0.16<br>[-2.19] | -0.18<br>[-2.41] | -0.20<br>[-2.50] | -0.25<br>[-3.21] | -0.15<br>[-1.87] | -0.21<br>[-3.02] |
| Anomaly 1       | 50.72<br>[15.77] |                  |                  |                  |                  |                  | 39.87<br>[11.06] |
| Anomaly 2       |                  | 39.69<br>[10.90] |                  |                  |                  |                  | 20.77<br>[3.47]  |
| Anomaly 3       |                  |                  | 36.63<br>[9.92]  |                  |                  |                  | 4.08<br>[0.68]   |
| Anomaly 4       |                  |                  |                  | 21.38<br>[4.56]  |                  |                  | 9.79<br>[2.14]   |
| Anomaly 5       |                  |                  |                  |                  | 33.58<br>[7.80]  |                  | 9.10<br>[2.06]   |
| Anomaly 6       |                  |                  |                  |                  |                  | 18.59<br>[4.29]  | -5.82<br>[-1.45] |
| mkt             | 11.68<br>[7.11]  | 11.56<br>[6.53]  | 11.84<br>[6.61]  | 12.33<br>[6.58]  | 12.47<br>[6.82]  | 12.35<br>[6.55]  | 10.99<br>[6.90]  |
| smb             | 26.83<br>[11.29] | 21.13<br>[7.96]  | 24.44<br>[9.30]  | 27.14<br>[9.91]  | 29.21<br>[11.06] | 29.80<br>[10.91] | 21.99<br>[8.95]  |
| hml             | 0.91<br>[0.28]   | -1.35<br>[-0.39] | -0.41<br>[-0.12] | 5.78<br>[1.59]   | 3.84<br>[1.08]   | 2.59<br>[0.69]   | -3.25<br>[-1.02] |
| rmw             | 26.23<br>[8.19]  | 23.70<br>[6.85]  | 22.98<br>[6.54]  | 26.21<br>[7.16]  | 26.79<br>[7.51]  | 26.96<br>[7.33]  | 24.55<br>[7.86]  |
| cma             | 24.70<br>[4.68]  | 42.71<br>[7.95]  | 43.89<br>[8.05]  | 37.83<br>[4.87]  | 38.58<br>[6.32]  | 47.57<br>[7.23]  | 6.16<br>[0.86]   |
| umd             | 5.36<br>[3.29]   | 3.09<br>[1.75]   | 2.74<br>[1.53]   | 6.12<br>[3.28]   | 4.75<br>[2.62]   | 4.26<br>[2.26]   | 4.08<br>[2.53]   |
| # months        | 732              | 728              | 728              | 732              | 732              | 713              | 709              |
| $\bar{R}^2(\%)$ | 54               | 46               | 45               | 39               | 42               | 39               | 57               |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CEB. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

## References

- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H. (1991). Production-based asset pricing and the link between stock returns and economic fluctuations. *Journal of Finance*, 46(1):209–237.
- Cooper, I. and Priestley, R. (2011). Real investment and risk dynamics. *Journal of Financial Economics*, 101(1):182–205.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Lettau, M. and Ludvigson, S. (2001). Consumption, aggregate wealth, and expected stock returns. *Journal of Finance*, 56(3):815–849.
- Lin, X. and Zhang, L. (2013). The investment manifesto. *Journal of Monetary Economics*, 60(3):351–366.

- Liu, L. X., Whited, T. M., and Zhang, L. (2009). Investment-based expected stock returns. *Journal of Political Economy*, 117(6):1105–1139.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Titman, S., Wei, K. J., and Xie, F. (2004). Capital investments and stock returns. *Journal of Financial and Quantitative Analysis*, 39(4):677–700.
- Xing, Y. (2008). Interpreting the value effect through the q-theory: An empirical investigation. *Review of Financial Studies*, 21(4):1767–1795.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.